Nile Nexus at ImageEval 2026 Shared Tasks: Single-Choice Reformulation and Confidence-Calibrated Refinement for Culturally Grounded Hallucination Detection

Sumaiya Zaman, Miftahul Jannat Rishta
Department of Computer Science and Engineering,
Chittagong University of Engineering and Technology, Bangladesh
{u2104110, u2104019}@student.cuet.ac.bd

Abstract

Visual hallucination detection in culturally grounded images has remained challenging due to fine-grained visual distinctions and knowledge-dependent statements. In the ImageEval2026 / ArabicNLP26 AynVQA Shared Task, we have participated in Task 1b by developing a system based on Qwen2.5-VL-7B-Instruct with 4-bit quantization. We have reformulated hallucination detection as a single-choice selection problem to enforce the exactly-one-correct-statement constraint and have introduced confidence-gated selective refinement using paraphrase ensemble voting. Our final system has achieved 0.942 accuracy, 0.058 CI, 0.942 Q+ accuracy and 0.971 Q− accuracy on the blind test set, ranking 6th in the English track; CFHR is structurally 0 under our single-choice formulation (see Section 5.1). Error analysis has revealed that the remaining failures have mostly stemmed from fine-grained cultural and visual grounding challenges rather than prediction inconsistency.

1 Introduction

Vision–language models (VLMs) have continued to accept statements unsupported by visual evidence, a failure mode known as hallucination (Ji et al., 2023; Rohrbach et al., 2018). This challenge has grown more pronounced for culturally specific concepts, as general-purpose VLMs have not captured regional knowledge uniformly.

The AynVQA Task 1b shared task, part of ImageEval 2026 / ArabicNLP26, has addressed this through culturally grounded hallucination detection: each instance has paired an image from an Arab country with three candidate statements, only one of which has been visually grounded. Systems have been scored using *Contrastive Instability* (CI), measuring incomplete predictions and CFHR, capturing cases where correct and false statements have both been accepted.

This paper has presented our submission, based on **Qwen2.5-VL-7B-Instruct** (Bai et al., 2025) in 4-bit precision. We have reformulated hallucination detection as single-choice selection, enforcing the exactly-one-correct constraint by design and have introduced confidence-gated selective refinement, re-evaluating low-confidence predictions through paraphrase-ensemble voting.

Our contributions have been:

- A single-choice reformulation that enforces the exactly-one-correct constraint by design, which structurally guarantees CFHR = 0 and reduces CI to a direct function of accuracy (see Section 5.1 for a discussion of this trade-off).
- A confidence-based selective refinement strategy that has improved reliability without additional training or external data.
- An evaluation showing improvement from 0.908 to 0.940 combined accuracy and a CI reduction from 0.092 to 0.060 on the English dev set, supported by category-level error analysis.

Implementation details are available at https://github.com/sumaiyaa110/-ImageEval-2026-Ayn-VQA-Task-1b.

2 Background

2.1 Task Overview

The ImageEval 2026 shared task (Abdaljalil et al., 2026) has evaluated culturally grounded multimodal reasoning by determining whether models can distinguish between statements that are visually supported and those that are hallucinated. We have participated in Task 1b: Hallucination Detection, English Track, where each instance has consisted of an image and three candidate statements, only one of which has been grounded in the image and the model has had to identify it.

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2.2 Dataset Description

The ImageEval 2026 dataset has covered culturally grounded samples from Arab countries across nine cultural categories and 31 subcategories, including food, clothing, traditions, objects and daily activities (Abdaljalil et al., 2026), with English and Modern Standard Arabic tracks. For Task 1b, each instance has followed the same one-image, three-statement structure described above. The dataset has provided training, development, development-test and blind-test splits, shown in Table 1.

Split	Samples	Purpose
Training	3,000	Not used (zero-shot system; see §3.7)
Development	500	Threshold selection, prompt refinement
Development-Test	500	Not scored in this submission
Test	1,000	Official evaluation
Total	**5,000**	

Table 1: Dataset distribution for ImageEval 2026 Task 1b.

2.3 Related Work

Vision-language models (VLMs) have achieved strong multimodal understanding; however, they have continued to suffer from hallucination, where generated information has not been grounded in visual evidence. Benchmarks such as CHAIR (Rohrbach et al., 2018) and POPE (Li et al., 2023) have evaluated visual hallucination and have shown that strong language generation has not always resulted in reliable visual grounding.

Recent research has also emphasized culturally grounded multimodal evaluation. The OASIS dataset (Alam et al., 2025) has introduced a multilingual benchmark for culturally grounded visual question answering, demonstrating the value of evaluating models beyond general visual concepts. ImageEval 2026 (Abdaljalil et al., 2026) has followed this direction, focusing on culturally aware multimodal reasoning across Arabic-speaking regions. Mousi et al. (Mousi et al., 2026b) have investigated counterfactual hallucination from semantically plausible but visually unsupported statements, while Mousi et al. (Mousi et al., 2026a) have analyzed instability between visual perception and textual reasoning.

Unlike prior benchmarks centered on general captioning and visual question answering, ImageEval 2026 has formulated hallucination detection as culturally grounded contrastive reasoning. Our work has addressed this by modeling the task as structured statement selection and introducing confidence-based refinement through multiple prompt variations, without additional training data.

3 System Overview

3.1 Task Reformulation

Our system has employed Qwen2.5-VL-7B-Instruct for Task 1b hallucination detection. We have reformulated the task as contrastive selection rather than independent per-statement True/False classification: given an image and three candidate statements, the model has identified the one best supported by the visual content. This has enforced the exactly-one-correct constraint and has reduced inconsistent predictions.

3.2 Model and Precision

To fit Kaggle GPU constraints (T4/P100), we have loaded the model in 4-bit NF4 precision via `bitsandbytes` with `float16` compute dtype. To preserve fine-grained visual features, such as woven textures and regional vessel shapes, we have raised the visual token budget to min_pixels $= 256 \times 28^2$ and max_pixels $= 2560 \times 28^2$. We have run batched inference (batch size 8) with left-padded tokenization.

3.3 Culture-Aware Prompting

Since AynVQA has featured culturally specific MENA imagery, we have designed prompts instructing the model to combine literal visual evidence with cultural or world knowledge when distinguishing candidates. The direct instruction prompt is given in full in Appendix A; the seven paraphrase variants used during selective refinement (§3.6) are released in full in the project's GitHub repository (linked in §1).

3.4 Pipeline

The pipeline has consisted of two stages, initial prediction and selective refinement, as shown in Figure 1.

3.5 Confidence Estimation

We have extracted the probability over the three candidates from the first generated token:

$$C = \max(P(s_1), P(s_2), P(s_3)) \quad (1)$$

3.6 Selective Refinement

Predictions with confidence below 0.85 have undergone refinement: we have re-evaluated them using

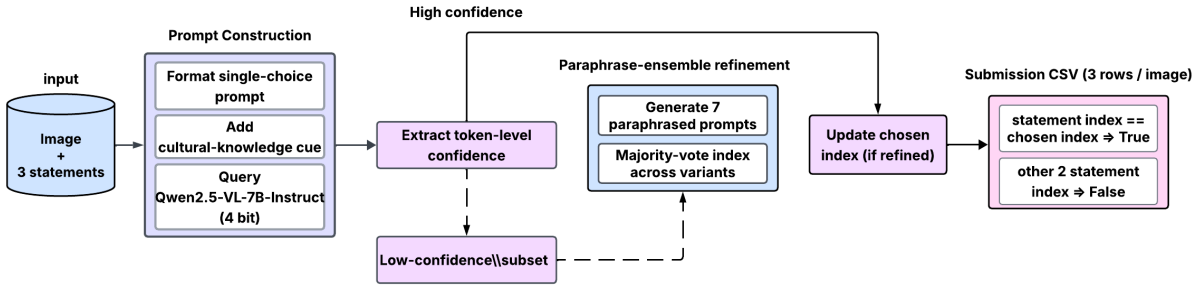


Figure 1: Proposed pipeline: single-choice VLM prompting, confidence-gated paraphrase-ensemble refinement for low-confidence items and expansion of the final index into per-statement True/False submission rows.

seven semantically equivalent prompt variants and have selected the final answer by majority vote:

$$\hat{s} = \arg\max_{i} \sum_{j=1}^{N} \mathbb{I}(s_j = i), \quad N = 7 \qquad (2)$$

The 0.85 threshold was selected by inspecting per-bin accuracy on the development set (Figure 2): predictions below this point showed markedly lower pre-refinement accuracy than predictions above it, indicating that low first-token confidence is a useful signal for which items benefit most from paraphrase-ensemble refinement. We do not report a formal calibration metric (e.g. ECE) for this threshold and refer to the procedure as confidence-*gated* rather than confidence-calibrated refinement.

3.7 Design Rationale

We have restricted refinement to uncertain samples to limit inference cost. The system has used no external training data, augmentation, or fine-tuning; gains have come entirely from prompt design, confidence gating and self-consistency refinement. The 3,000-sample training split was not used because our approach targets a zero-shot, fine-tuning-free setting under Kaggle's free-tier GPU constraints (T4/P100), where full or parameter-efficient fine-tuning of a 7B VLM was not practically feasible within the compute and time budget available to us; we treat this as a design constraint rather than an oversight, and view supervised fine-tuning on the training split as a direction for future work rather than a component of the current system.

4 Experimental Setup

4.1 Task and Data

We have evaluated our system on the ImageEval 2026 Task 1b English track. The development split has supported analysis, confidence-threshold selection and prompt refinement; the final submission has run on the 1,000-item blind test split. Each input has consisted of an image and three statements, one grounded in the image. The development-test split was not scored in this submission; the threshold and prompts reported here were selected and evaluated on the same development split, and we flag this as a limitation in Section 6.

4.2 Model and Inference Configuration

We have used Qwen2.5-VL-7B-Instruct with 4-bit NF4 quantization, FP16 computation, batch size 8 and a refinement threshold of 0.85. Detailed inference configurations are provided in Appendix B.

4.3 Evaluation Metrics

The evaluation has used five metrics: CI (ranking metric), combined accuracy, CFHR, Q+ accuracy and Q− accuracy. Q+ and Q− accuracy have measured grounded-statement identification and hallucination rejection. Combined accuracy has required all three predictions correct. CI has measured the share of at-least-one-correct instances not fully correct, so a system that trivially labels every statement false does not achieve a low CI; CFHR, the share also accepting a hallucination. Under our single-choice formulation, exactly one statement is ever predicted True per instance, so CFHR (accepting a hallucination alongside the correct statement) is structurally 0 by construction rather than an empirical finding, and CI reduces to $1 -$ accuracy. We report both for completeness and comparability with per-statement True/False systems, but interpret them as properties of the reformulation rather than independent results.

5 Results and Analysis

5.1 Comparison of Inference Strategies

Table 2 has compared direct single-choice prediction, chain-of-thought (CoT) prompting, and confidence-gated selective refinement on the English development set. Single-choice prediction has achieved 0.908 accuracy and 0.092 CI. CoT has degraded performance to 0.588 accuracy and 0.412 CI; our CoT harness allows up to 40 generated tokens (versus 8 for the direct-prompt condition) and parses the answer from the last digit in the output (Appendix B). The larger budget rules out simple truncation as the primary cause; the drop more likely reflects the last-digit parsing rule misreading reasoning traces that reference multiple digits before settling on a final answer, or the additional reasoning itself introducing errors. We do not treat this comparison as a clean ablation of CoT prompting in general (see Section 6). Confidence-gated refinement has achieved the best results, 0.940 accuracy and 0.060 CI, confirming that selective refinement has increased reliability. As noted in Section 4, CFHR and CI in Table 2 follow directly from accuracy under our formulation.

5.2 Analysis of Confidence-Based Refinement

The refinement strategy has applied additional inference only to uncertain cases. On the 500-sample development set — used for threshold selection and the analysis in this section — 209 (41.8%) low-confidence predictions were refined using seven paraphrased prompts and majority voting, correcting 20 errors while introducing 4 new mistakes, a net gain of 16 correct predictions.

On the 1,000-item blind test set, the same refinement procedure was applied at the fixed 0.85 threshold, correcting 42 errors while introducing 6 new mistakes, a net gain of 36 correct predictions. Figure 2 shows this test-set effect broken down by initial confidence bin: refinement produced the largest gains in the lowest-confidence bins, while high-confidence predictions were largely unchanged.

5.3 Error Analysis

Prediction accuracy differs significantly across the three possible grounded-statement positions (91.5%, 96.5% and 94.5% for positions 0, 1 and 2 respectively; $\chi^2(2, N = 1000) = 7.88$, $p = 0.020$; see Appendix C). A pairwise comparison shows position 0 is significantly worse than position 1 (Fisher's exact test, $p = 0.008$, surviving Bonferroni correction for three comparisons), while the other two pairwise differences are not significant. We therefore do not claim the system is free of positional bias; the effect is smaller than the overall accuracy gain from refinement but is not attributable to chance.

5.3.1 Qualitative Error Analysis

Remaining errors have mainly occurred in Geography, Buildings & Landmarks. Failures have resulted from fine-grained visual similarity, cultural knowledge not directly observable from images and multiple visually plausible statements. This has indicated that the proposed single-choice formulation has resolved structural hallucination, while visual-cultural grounding has remained challenging.

6 Conclusion

We have presented our submission to Task 1b of the AynVQA shared task at ImageEval2026 / ArabicNLP26. We have reformulated multi-statement hallucination evaluation into a single-choice framework that has enforced the exactly-one-true constraint and has removed structural prediction inconsistencies. Built on 4-bit quantized `Qwen2.5-VL-7B-Instruct` with confidence-gated paraphrase-ensemble voting, our approach has achieved 0.942 accuracy and 0.058 CI on the blind test set, ranking 6th; CFHR remains 0 as a structural property of the single-choice reformulation rather than an independently achieved result. Remaining errors have stemmed mainly from fine-grained visual-cultural distinctions rather than format-level inconsistencies.

Limitations

Several limitations remain. 4-bit NF4 quantization may have introduced minor alignment loss. Errors have mainly occurred in geography and landmarks, where external knowledge is needed beyond visual content. The paraphrase ensemble has improved accuracy but increased inference cost, and the single-choice setup limits extension to unconstrained scenarios. The 0.85 refinement threshold and prompt variants were selected and evaluated on the same development split used to report the accuracy gains in Table 2, without an independent devtest estimate, so the reported improvement may be optimistic relative to unseen data. The chain-of-thought comparison in Section 5.1 is confounded by last-digit parsing over a 40-token budget, which

Method	Accuracy	CI	CFHR	Q+	Q-
Single-choice Prediction	0.908	0.092	0.000	0.908	0.954
Chain-of-Thought Prompting	0.588	0.412	0.000	0.588	0.794
Confidence-Gated Refinement	**0.940**	**0.060**	**0.000**	**0.940**	**0.970**
Final Submission (Test Set)	**0.942**	**0.058**	**0.000**	**0.942**	**0.971**

Table 2: Comparison of inference strategies on the ImageEval 2026 Task 1b English development set, together with the official test-set leaderboard performance of the final submission. CFHR is 0 for every row by construction of the single-choice format (Section 4); Q+ and Q− are derived directly from combined accuracy.

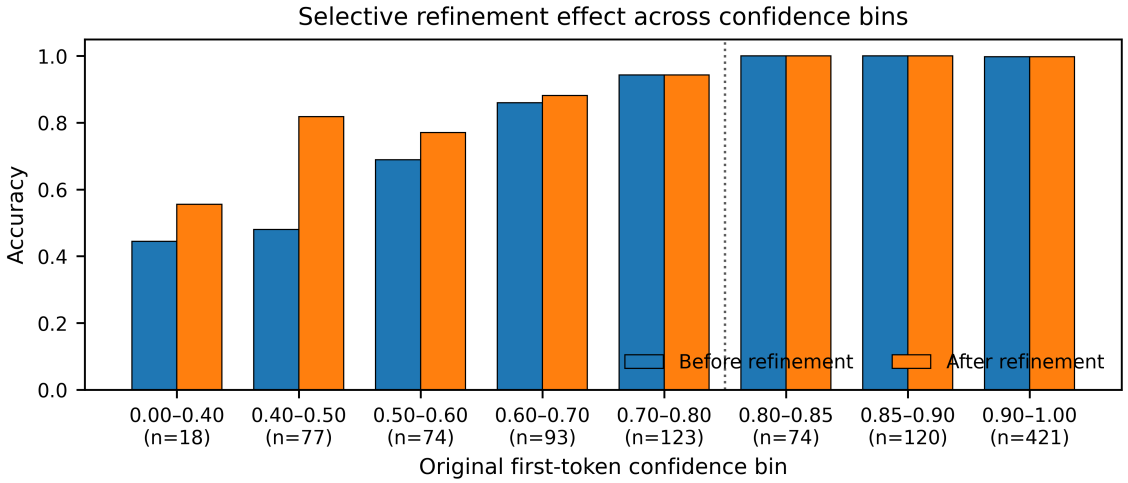


Figure 2: Effect of selective refinement across initial confidence bins on the English test set. Refinement has produced the largest gains for low-confidence predictions, with high-confidence predictions largely unchanged; values in parentheses indicate instance counts per bin.

may misparse traces that mention multiple digits before a final answer, so we do not treat it as a clean test of CoT prompting in general. Finally, we compare against neither the organizers' baseline nor an alternative backbone, and do not use the 3,000-item training split, as our system targets a zero-shot, fine-tuning-free setting under free-tier GPU constraints (Section 3).

Ethical Statement

This work has used publicly available AynVQA benchmark data, following dataset guidelines. We have collected or processed no personal data.

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A Prompts

A.1 Direct Instruction Prompt

```
You will see an image and three statements about it. Exactly ONE statement is actually correct for this image; the other two are plausible-sounding but wrong. Judge using BOTH what is literally visible in the image AND any relevant real-world or cultural knowledge needed to tell the options apart (e.g. the specific type of object, dish, or vessel shown, not just its general appearance). Reply with ONLY the number (0, 1, or 2) of the one correct statement. Do not explain, just output the digit.
```

The seven paraphrase variants used during selective refinement (§3.6) are semantically equivalent restatements of this prompt and are released in full in the project's GitHub repository (§1).

B Detailed Experimental Configuration

Table 3 has provided the detailed inference configuration used in our experiments, including the chain-of-thought harness discussed in Section 5.1.

Configuration	Value
Model	Qwen2.5-VL-7B-Instruct
Quantization	4-bit NF4
Computation Type	FP16
Maximum Generated Tokens (direct)	8
Maximum Generated Tokens (CoT)	40
Batch Size	8
Refinement Threshold	0.85
CoT Answer Parsing	Last digit in output

Table 3: Experimental configuration of the proposed system. The 8-token generation cap applies to the direct single-choice condition; the chain-of-thought condition (Section 5.1) was allowed up to 40 tokens, with the answer parsed from the last digit in the output.

C Grounded-Statement Position Analysis

To verify that the model has not relied on statement ordering, we have analyzed the prediction distribution across the three possible grounded-statement positions. Figure 3 shows the corresponding confusion matrix on the English test set.

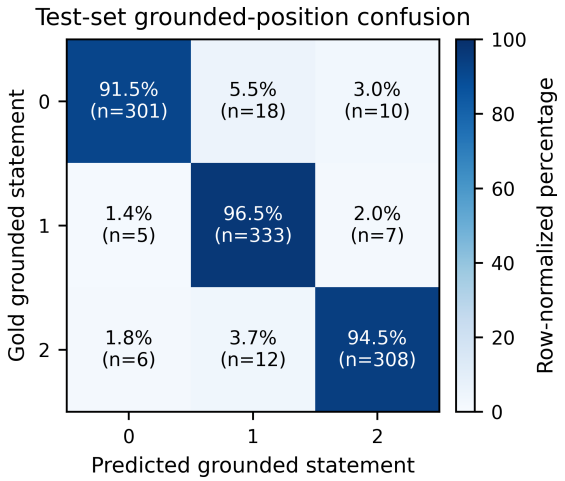


Figure 3: Confusion matrix of predicted grounded-statement position on the English test set, normalized by gold position, with raw counts in parentheses.

A chi-square test of homogeneity on the diagonal (correct/incorrect) counts per position $(329, 345, 326$ instances at positions $0, 1, 2)$ gives $\chi^2(2) = 7.88$, $p = 0.020$, rejecting equal accuracy across positions. Pairwise Fisher's exact tests give $p = 0.008$ (position 0 vs. 1), $p = 0.168$ (0 vs. 2), and $p = 0.262$ (1 vs. 2); only the first survives a Bonferroni-corrected threshold of $\alpha = 0.0167$. These tests were computed with `scipy.stats.chi2_contingency` and `scipy.stats.fisher_exact` on the diagonal counts above.